

EN pin connection for TPSM863252RDXR

Regarding the TPSM863252RDXR, EN pin, and external power source from 7 to 15 VDC, do i should connect EN pin directly on VIN ?

I'll verify the TPSM863252 EN specifications against the TI datasheet, then answer with the practical connection recommendation for a 7–15 V input design.

No — **do not connect EN directly to VIN** when VIN can be **7–15 VDC**.

For the **TPSM863252RDXR**, the **EN pin absolute maximum is 6 V**, while VIN may go up to 15 V. Direct EN-to-VIN would overvoltage the EN pin. TI also says EN may be driven by a divider from VIN or by a voltage below 5.5 V, and that EN must not exceed 6 V. [TI](#) [TI](#)

Use one of these:

Best simple solution: resistor from VIN to EN

The IC has an internal **2 MΩ pulldown** on EN. TI explicitly allows using only a **top resistor from VIN to EN**, using the internal 2 MΩ as the lower divider resistor. [TI](#)

For VIN = **7–15 V**, a practical value is:

VIN — 4.7 MΩ — EN
EN internally pulled down by 2 MΩ

Approximate EN voltage:

$$V_{EN} = V_{IN} \times 2 \text{ M}\Omega / (4.7 \text{ M}\Omega + 2 \text{ M}\Omega)$$

So:

At VIN = 7 V → VEN ≈ 2.09 V
At VIN = 15 V → VEN ≈ 4.48 V

That is safely above the EN turn-on threshold, which is max **1.25 V**, and below the 6 V maximum. [TI](#)

More robust solution: external divider

Use this if the environment is noisy or the EN trace is long:

VIN — 1 MΩ — EN — 270 kΩ — GND

Remember the internal 2 MΩ pulldown is in parallel with the 270 kΩ, so the real lower resistance is about 238 kΩ.

Approximate EN voltage:

At VIN = 7 V → VEN ≈ 1.35 V
At VIN = 15 V → VEN ≈ 2.88 V

This is also valid and lower impedance than the 4.7 M Ω -only solution.

Recommendation for your case

For a **7–15 V external source**, I would use:

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VIN — 4.7 M $\Omega$  — EN
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or, for better noise immunity:

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VIN — 1 M $\Omega$  — EN — 270 k $\Omega$  — PGND
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Do **not** tie EN directly to VIN.